Lecture 10: LTI Resistive Circuits and Equivalent One-Ports

EE201 – Circuit Theory I

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current voltage a part of a circuit (box N)

Review: equations from the graph alone

	incidence / mesh	**fundamental cutset / loop**
KCL	$A\mathbf{i} = \mathbf{0}$	$Q\mathbf{i} = \mathbf{0}$
KVL	$M\mathbf{v} = \mathbf{0}$	$B\mathbf{v} = \mathbf{0}$
$n-1$ voltages	node voltages $\mathbf{e}$	tree voltages $\mathbf{v}_T$
$b-n+1$ currents	mesh currents $\mathbf{i}_M$	co-tree currents $\mathbf{i}_C$
works for	M: planar, nonseparable	any connected graph

KCL and KVL depend only on the graph: they hold for any elements.

A circuit has $2b$ unknowns: b more equations come from the elements.

Why this lecture?

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  - substitution theorem, equivalent one-ports
  - series and parallel connections, source transformation

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- **This lecture — transform:** replace a part by a source, or by a simpler part with the same terminal behaviour
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  - series and parallel connections, source transformation
- **Later — the circuit as it is:** node voltage and mesh current analysis

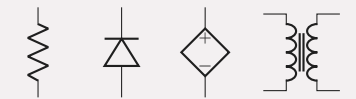
Roadmap

1. **LTI resistive circuits** — what the words mean, and what we gain
2. **A first example** — simplify a circuit step by step
3. **Substitution theorem** — replace a part of a circuit by a source
4. **One-ports and equivalence** — the same i–v characteristic at the port
5. **Series and parallel** — combine resistors and sources
6. **One source and one resistor** — exchange a voltage source and a current source

Resistive and dynamic elements

Resistive element: no memory

Its law relates v and i at the same instant, algebraically.

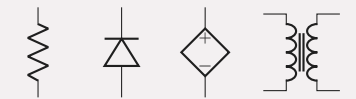


later: the ideal op-amp

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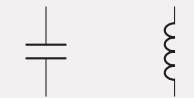
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Dynamic element: memory

Its law has a derivative.

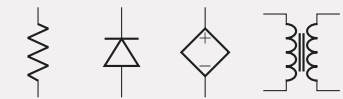


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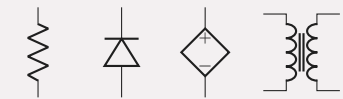
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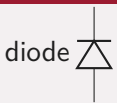
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"Resistive" means memoryless, not only "a resistor".

An element is linear if its law is linear in v, i; time-invariant if its law does not change with time.

Three questions about a circuit

Linear?

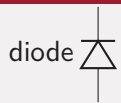


Nonlinear if it contains at least one nonlinear element, e.g. a diode.

Otherwise **linear**.

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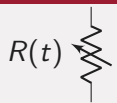
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Time-invariant?

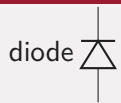


Time-varying if it contains at least one time-varying element, e.g. $R(t)$.

Otherwise **time-invariant**.

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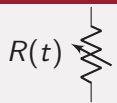
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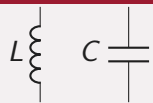
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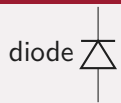


Dynamic if it contains at least one dynamic element: an L or a C.

Otherwise **resistive**.

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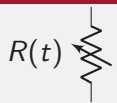
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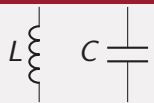
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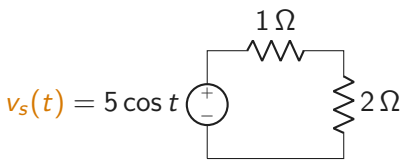
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LTI resistive circuit = linear + time-invariant + resistive:

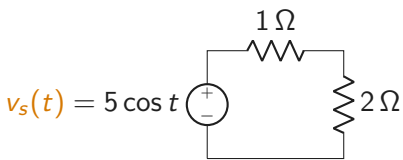
apart from the independent sources, every element is linear, time-invariant and resistive.

Independent sources are inputs



The source varies in time. Is the circuit time-varying?

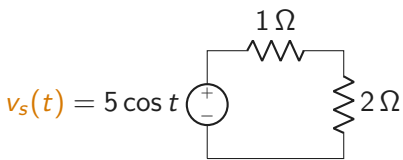
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No. An independent source is an external input: a given signal of time.

The three questions are about the circuit that the inputs drive, not about the inputs. This is an LTI resistive circuit.

What varies in time?

In every circuit $v(t)$ and $i(t)$ are functions of time. The question is whether the element law changes with time.

Time-invariant resistor

$$v(t) = R\, i(t)$$

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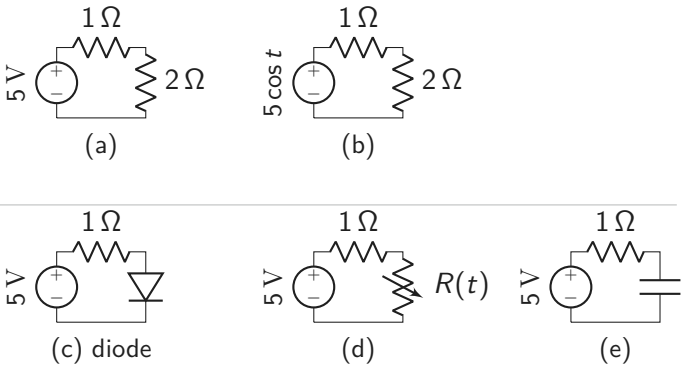
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Linear or nonlinear is also a property of the element law: $v = Ri$ is linear, a diode law is not.

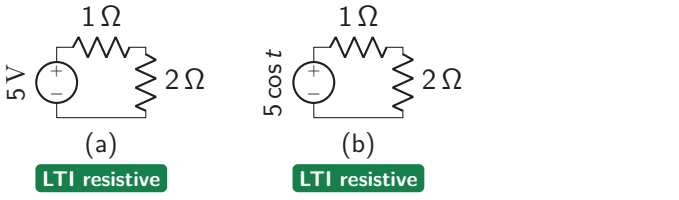
Which circuit is which?



A source, a $1\,\Omega$ resistor and one more element. In (d), $R(t) = 2 + \sin t\ \Omega$.

Linear? Time-invariant? Resistive?

Which circuit is which?



The input does not change the class: (a) and (b) are both LTI resistive.

Each of (c), (d), (e) breaks exactly one of the three properties.

What we gain

Always — no assumption on the elements

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Because the circuit is LTI resistive

Every element equation is linear and algebraic, with no derivative:

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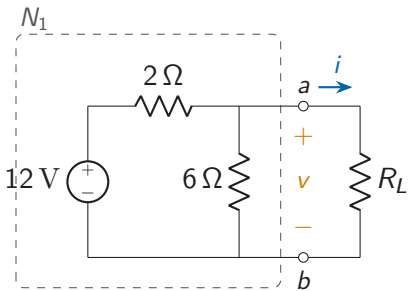
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- Time enters only through the values of the independent sources.
- So at each instant the circuit is solved like a circuit with constant sources: $2b$ linear algebraic equations.

Roadmap

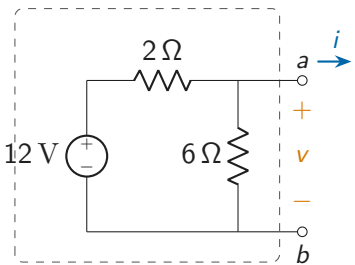
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A first example: what does R_L see?



R_L (the load) can be any resistive element.
Find v and i.

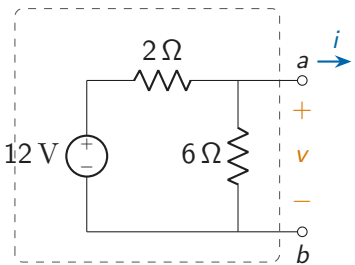
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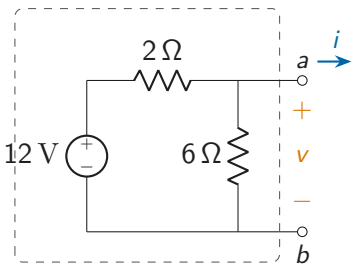


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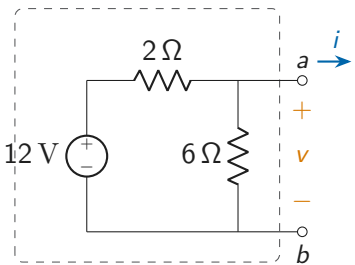
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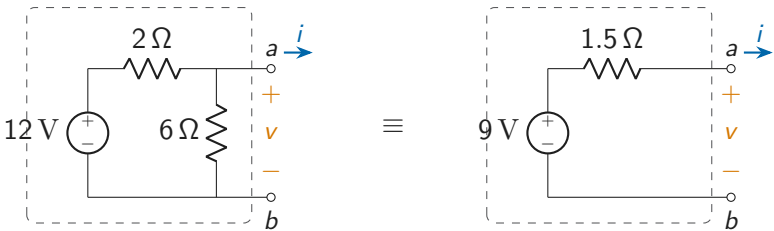
Solve for v: $$v = 9 - 1.5\,i$$

Read the result as a circuit

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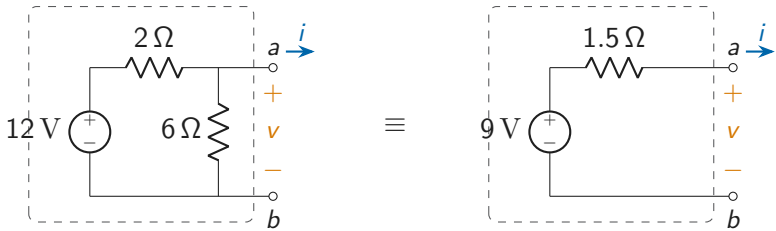
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The right one obeys $v = 9 - 1.5\,i$ too (KVL): the same law.

Read the result as a circuit

$$v = 9 - 1.5\,i$$



The right one obeys $v = 9 - 1.5\,i$ too (KVL): the same law.

Any load: $v = R_L i$, so $i = \dfrac{9}{1.5 + R_L}$. For $R_L = 3\,\Omega$: $i = 2\,\mathrm{A}$, $v = 6\,\mathrm{V}$.

What did we just do?

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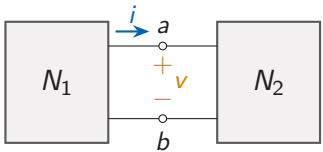
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question	**answered by**
May a part be replaced by a source?	substitution theorem
When may a part be replaced by a simpler part?	equivalent one-ports
Which simple replacements are there?	series, parallel, source transformation

Roadmap

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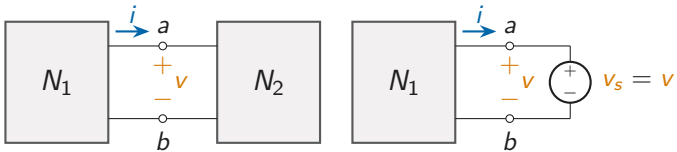
The substitution theorem



(I)

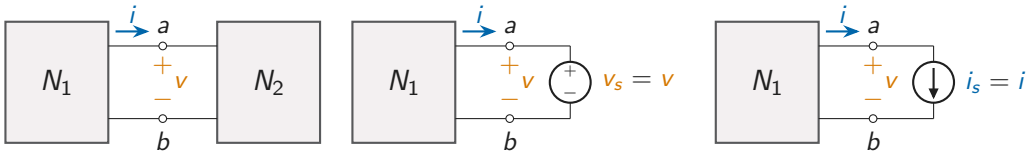
A circuit cut into two parts, joined at the terminals a, b.
i: current from N_1 into N_2 v: voltage between a and b

The substitution theorem



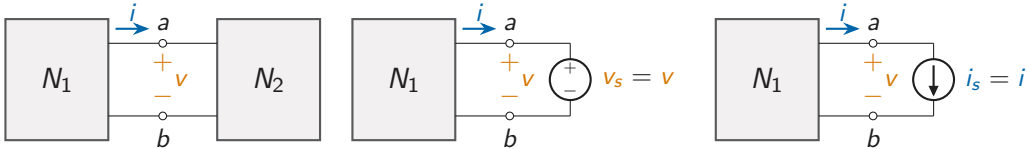
Replace N_2 by a voltage source with $v_s(t) = v(t)$ ...

The substitution theorem



... or by a current source with $i_s(t) = i(t)$.

The substitution theorem



Substitution theorem

No branch voltage or branch current inside N_1 changes. An observer inside N_1 cannot tell (I), (II) and (III) apart.

Condition: each circuit has a unique solution.

A unique solution

A circuit with n nodes and b branches has $2b$ unknowns, $v_1, \ldots, v_b$ and $i_1, \ldots, i_b$, and $2b$ equations:

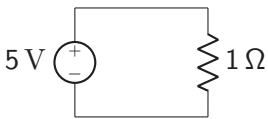
KCL	$A\mathbf{i} = 0$	$n-1$
KVL	$B\mathbf{v} = 0$	$b-n+1$
element laws	one for each branch	b

Definition

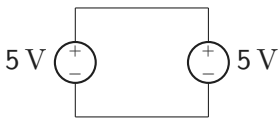
The circuit has a unique solution if exactly one $(\mathbf{v}, \mathbf{i})$ satisfies all $2b$ equations.

How do we check it? For an LTI resistive circuit: see the next slides.

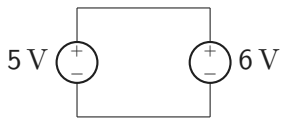
A unique solution? Three small circuits



(a)

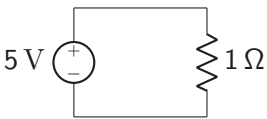


(b)



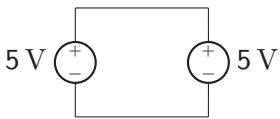
(c)

A unique solution? Three small circuits



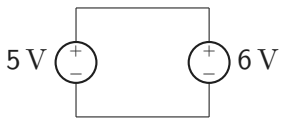
(a)

✓ unique: $i = 5\,\mathrm{A}$



(b)

× many: any current



(c)

× none: KVL fails

A unique solution: in matrix form

Collect the $2b$ unknowns in one vector $x = (v_1, \ldots, v_b, i_1, \ldots, i_b)$. In an LTI resistive circuit all $2b$ equations are linear: $T\,x = u$.

T, the coefficient matrix: one row per equation, holding the numbers that multiply the unknowns. u: the source values.

The solution is unique exactly when T is invertible: $\det T \neq 0$.

Branch 1: the left source; branch 2: the right element; both oriented from top to bottom.

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(a) $5\,\text{V}$  $1\,\Omega$

$$\begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 5 \\ 0 \end{bmatrix}$$

KCL
KVL
source 1
resistor 2

$\det T = -1$: ✓ unique

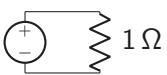
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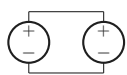
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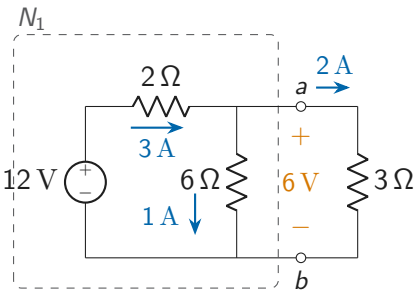
(b) $5\,\mathrm{V}$  $5\,\mathrm{V}$

$$\begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 5 \\ 5 \end{bmatrix} \begin{matrix} \text{KCL} \\ \text{KVL} \\ \text{source 1} \\ \text{source 2} \end{matrix}$$

$\det T = 0$: × i_1, i_2 appear only in KCL

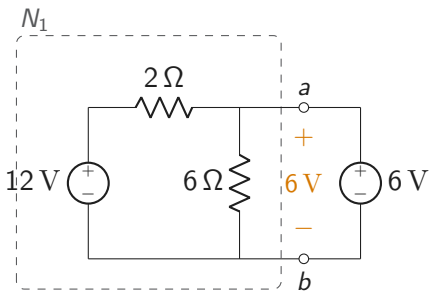
Branch 1: the left source; branch 2: the right element; both oriented from top to bottom.

(I) and (II): an example



(I) The first example with $R_L = 3\,\Omega$:
$v = 6\,\mathrm{V}$, $i = 2\,\mathrm{A}$; $3\,\mathrm{A}$ in the $2\,\Omega$, $1\,\mathrm{A}$ in the $6\,\Omega$.

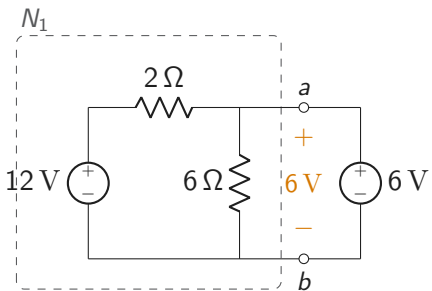
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(II) Replace R_L by a $6\,\mathrm{V}$ source. What are the values now?

(I) and (II): an example



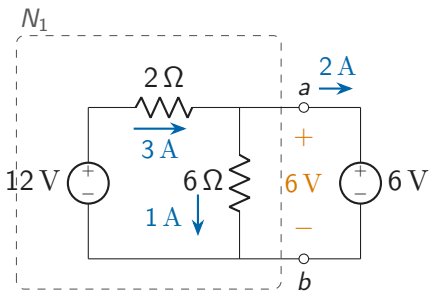
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KCL at a with $v = 6\,\mathrm{V}$:

$$\frac{12-6}{2} = \frac{6}{6} + i \;\Rightarrow\; i = 2\,\mathrm{A}$$

(I) and (II): an example



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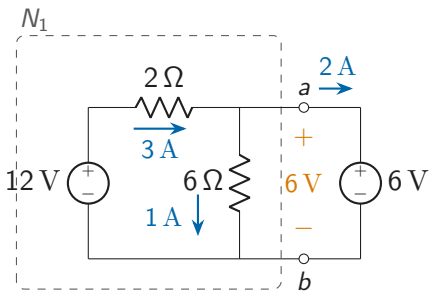
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$3\,\mathrm{A}$ in the $2\,\Omega$, $1\,\mathrm{A}$ in the $6\,\Omega$: nothing changed. ✓

(I) and (II): an example



(I) The first example with $R_L = 3\,\Omega$: $v = 6\,\mathrm{V}$, $i = 2\,\mathrm{A}$; $3\,\mathrm{A}$ in the $2\,\Omega$, $1\,\mathrm{A}$ in the $6\,\Omega$.

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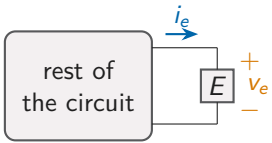
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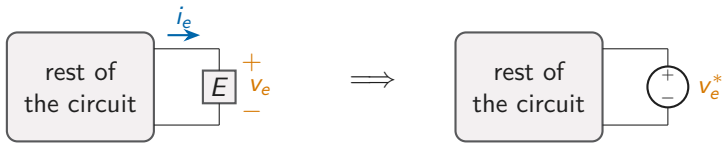
In (I) we solved the same equation: there too $v = 6\,\mathrm{V}$.

(I) and (II): one element first



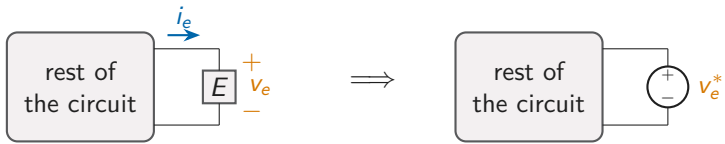
Any element E of a circuit; in the solution of (I) it has v_e^* and i_e^*.

(I) and (II): one element first



Replace only E by a voltage source. Assume both circuits have a unique solution.

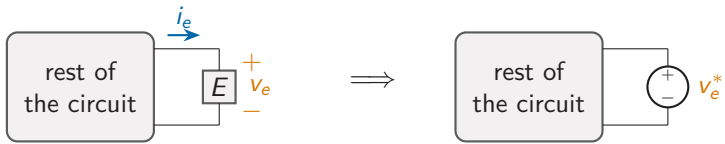
(I) and (II): one element first



Replace only E by a voltage source. Assume both circuits have a unique solution.

	(I)
KCL	$A\,\mathbf{i} = 0$
KVL	$B\,\mathbf{v} = 0$
other $b-1$ elements	their laws
element E	the law of E

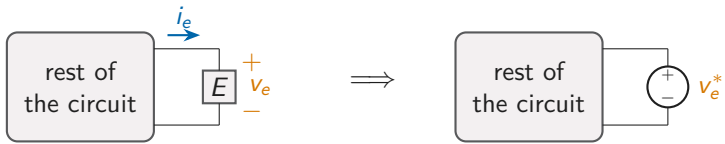
(I) and (II): one element first



Replace only E by a voltage source. Assume both circuits have a unique solution.

	(I)	**(II)**
KCL	$A\,\mathbf{i} = 0$	the same (same graph)
KVL	$B\,\mathbf{v} = 0$	the same (same graph)
other $b-1$ elements	their laws	the same
element E	the law of E	$v_e = v_e^*$

(I) and (II): one element first



Replace only E by a voltage source. Assume both circuits have a unique solution.

	(I)	**(II)**
KCL	$A\,\mathbf{i} = 0$	the same (same graph)
KVL	$B\,\mathbf{v} = 0$	the same (same graph)
other $b-1$ elements	their laws	the same
element E	the law of E	$v_e = v_e^*$

- The solution of (I), $(\mathbf{v}^*, \mathbf{i}^*)$, satisfies every equation of (I), the law of E included.

(I) and (II): one element, check the old solution

Does $(\mathbf{v}^*, \mathbf{i}^*)$ also satisfy the equations of (II)?

	(I)	(II)
KCL	$A\mathbf{i} = 0$	the same (same graph)
KVL	$B\mathbf{v} = 0$	the same (same graph)
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(I) and (II): one element, check the old solution

Does $(\mathbf{v}^*, \mathbf{i}^*)$ also satisfy the equations of (II)?

	(I)	(II)	$(\mathbf{v}^*, \mathbf{i}^*)$ in (II)?
KCL	$A\,\mathbf{i} = 0$	the same (same graph)	✓ as in (I)
KVL	$B\,\mathbf{v} = 0$	the same (same graph)	✓ as in (I)
other $b-1$ elements	their laws	the same	✓ as in (I)
element E	the law of E	$v_e = v_e^*$	✓ $v_e^* = v_e^*$

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	(I)	(II)	$(\mathbf{v}^*, \mathbf{i}^*)$ in (II)?
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element E	the law of E	$v_e = v_e^*$	✓ $v_e^* = v_e^*$

- A voltage source takes any current, so $i_e = i_e^*$ is allowed.

(I) and (II): one element, check the old solution

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- A voltage source takes any current, so $i_e = i_e^*$ is allowed.
- So $(\mathbf{v}^*, \mathbf{i}^*)$ solves (II); it has only one solution, so nothing changes, also $i_e = i_e^*$.

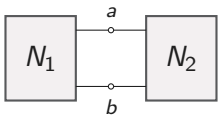
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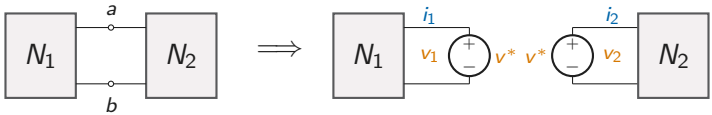
- A voltage source takes any current, so $i_e = i_e^*$ is allowed.
- So $(\mathbf{v}^*, \mathbf{i}^*)$ solves (II); it has only one solution, so nothing changes, also $i_e = i_e^*$.
- Uniqueness of (I) alone is not enough: in (a) of the three small circuits, replacing the $1\,\Omega$ by its voltage $5\,\mathrm{V}$ gives (b).

(I) and (II): divide N_1 and N_2



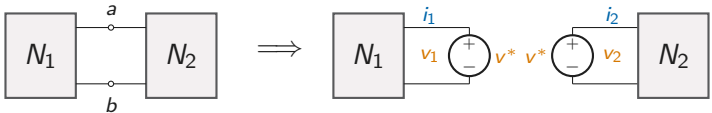
(I): N_1 and N_2 joined at a, b; the solution has v^*, i^* at the port.

(I) and (II): divide N_1 and N_2



Divide them and give each part its own source. Assume a unique solution.

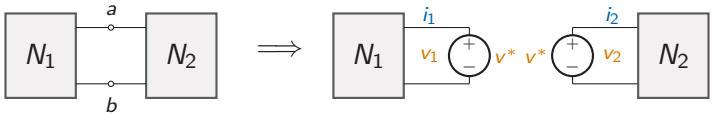
(I) and (II): divide N_1 and N_2



Divide them and give each part its own source. Assume a unique solution.

	(I)	**divided**
inside N_1	KCL, KVL, element laws	the same
inside N_2	KCL, KVL, element laws	the same
the connection	$v_1 = v_2,\ \ i_1 = i_2$	$v_1 = v^*,\ \ v_2 = v^*$

(I) and (II): divide N_1 and N_2

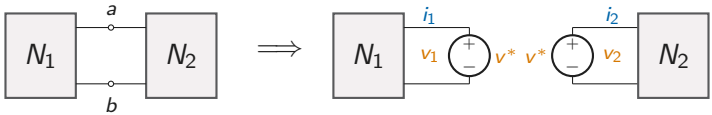


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- The solution of (I) satisfies all these equations, since $v_1 = v_2 = v^*$. ✓

(I) and (II): divide N_1 and N_2

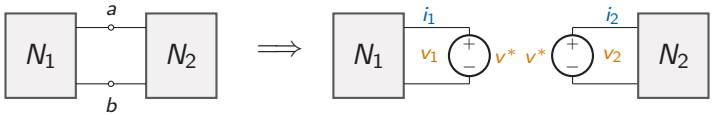


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- One solution, so this is it: nothing changes in either part.

(I) and (II): divide N_1 and N_2

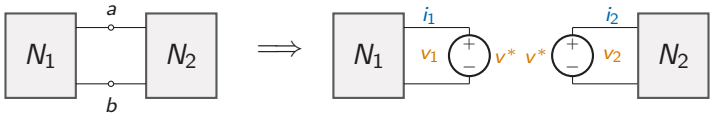


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- The left part is exactly (II): nothing inside N_1 changes, also $i_1 = i^*$.

(I) and (II): divide N_1 and N_2

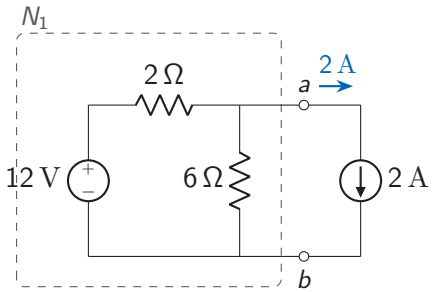


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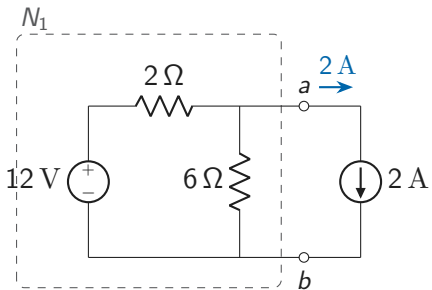
- The solution of (I) satisfies all these equations, since $v_1 = v_2 = v^*$. ✓
- One solution, so this is it: nothing changes in either part.
- The left part is exactly (II): nothing inside N_1 changes, also $i_1 = i^*$.
- N_2 is not needed at all: it can be anything.

(I) and (III): an example



(III) Replace R_L by a $2\,\mathrm{A}$ source. What are the values now?

(I) and (III): an example

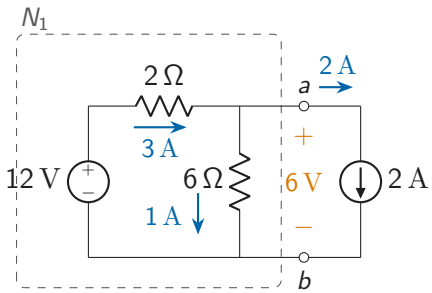


(III) Replace R_L by a $2\,\mathrm{A}$ source. What are the values now?

KCL at a with $i = 2\,\mathrm{A}$:

$$\frac{12 - v}{2} = \frac{v}{6} + 2 \;\Rightarrow\; v = 6\,\mathrm{V}$$

(I) and (III): an example



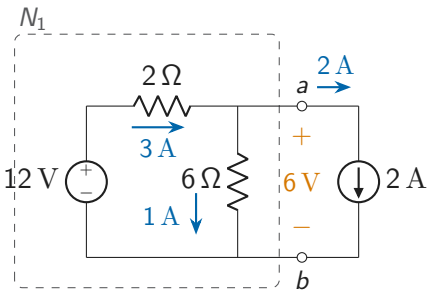
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KCL at a with $i = 2\,\mathrm{A}$:

$$\frac{12 - v}{2} = \frac{v}{6} + 2 \;\Rightarrow\; v = 6\,\mathrm{V}$$

$3\,\mathrm{A}$ in the $2\,\Omega$, $1\,\mathrm{A}$ in the $6\,\Omega$: nothing changed. ✓

(I) and (III): an example



(III) Replace R_L by a $2\,\mathrm{A}$ source. What are the values now?

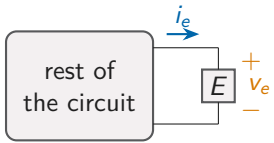
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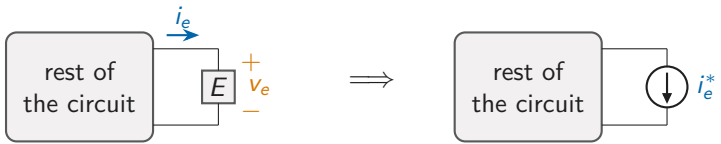
In (I) this equation holds too: there $i = 2\,\mathrm{A}$.

(I) and (III): one element first



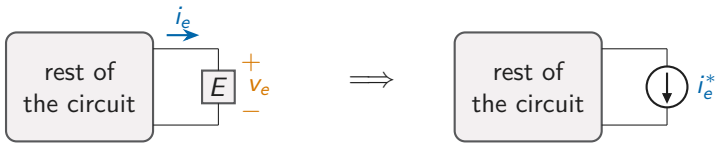
Any element E of a circuit; in the solution of (I) it has v_e^* and i_e^*.

(I) and (III): one element first



Replace only E by a current source. Assume both circuits have a unique solution.

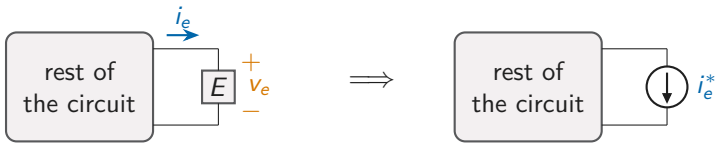
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Replace only E by a current source. Assume both circuits have a unique solution.

	(I)
KCL	$A\,\mathbf{i} = 0$
KVL	$B\,\mathbf{v} = 0$
other $b-1$ elements	their laws
element E	the law of E

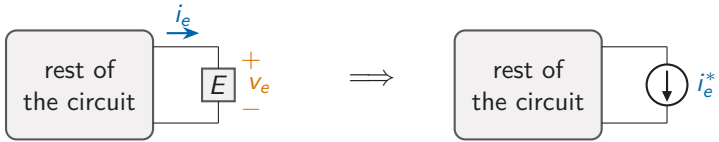
(I) and (III): one element first



Replace only E by a current source. Assume both circuits have a unique solution.

	(I)	**(III)**
KCL	$A\,\mathbf{i} = 0$	the same (same graph)
KVL	$B\,\mathbf{v} = 0$	the same (same graph)
other $b-1$ elements	their laws	the same
element E	the law of E	$i_e = i_e^*$

(I) and (III): one element first



Replace only E by a current source. Assume both circuits have a unique solution.

	(I)	(III)
KCL	$A\,\mathbf{i} = 0$	the same (same graph)
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element E	the law of E	$i_e = i_e^*$

- The solution of (I), $(\mathbf{v}^*, \mathbf{i}^*)$, satisfies every equation of (I), the law of E included.

(I) and (III): one element, check the old solution

Does $(\mathbf{v}^*, \mathbf{i}^*)$ also satisfy the equations of (III)?

	(I)	(III)
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(I) and (III): one element, check the old solution

Does $(\mathbf{v}^*, \mathbf{i}^*)$ also satisfy the equations of (III)?

	(I)	(III)	$(\mathbf{v}^*, \mathbf{i}^*)$ in (III)?
KCL	$A\,\mathbf{i} = 0$	the same (same graph)	✓ as in (I)
KVL	$B\,\mathbf{v} = 0$	the same (same graph)	✓ as in (I)
other $b-1$ elements	their laws	the same	✓ as in (I)
element E	the law of E	$i_e = i_e^*$	✓ $i_e^* = i_e^*$

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- A current source takes any voltage, so $v_e = v_e^*$ is allowed.

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- A current source takes any voltage, so $v_e = v_e^*$ is allowed.
- So $(\mathbf{v}^*, \mathbf{i}^*)$ solves (III); it has only one solution, so nothing changes, also $v_e = v_e^*$.

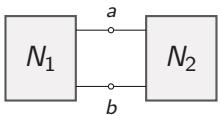
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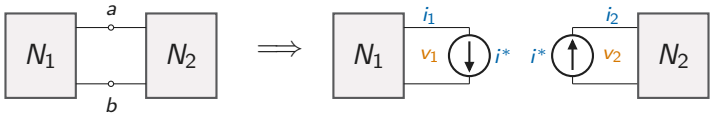
- A current source takes any voltage, so $v_e = v_e^*$ is allowed.
- So $(\mathbf{v}^*, \mathbf{i}^*)$ solves (III); it has only one solution, so nothing changes, also $v_e = v_e^*$.
- Uniqueness of (I) alone is not enough: a $2\,\mathrm{A}$ source with a $1\,\Omega$ becomes two $2\,\mathrm{A}$ sources in series.

(I) and (III): divide N_1 and N_2



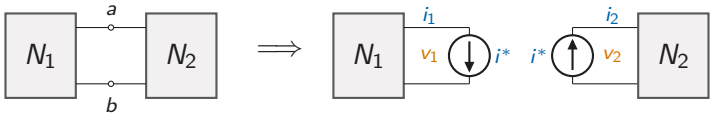
(I): N_1 and N_2 joined at a, b; the solution has v^*, i^* at the port.

(I) and (III): divide N_1 and N_2



Divide them and give each part its own source. Assume a unique solution.

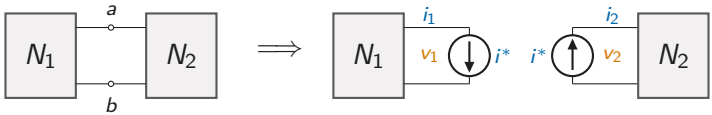
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inside N_1	KCL, KVL, element laws	the same
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the connection	$v_1 = v_2, \;\; i_1 = i_2$	$i_1 = i^*, \;\; i_2 = i^*$

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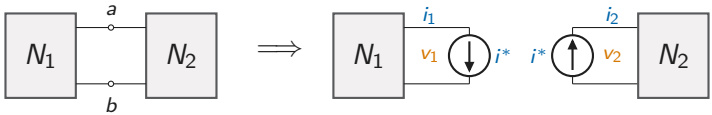


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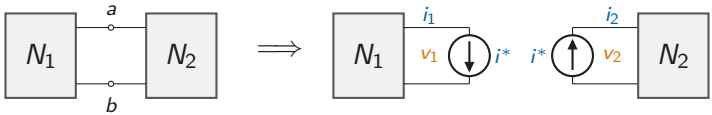


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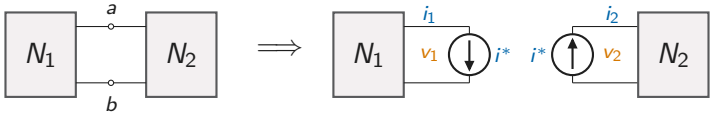


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- N_2 is not needed at all: it can be anything.

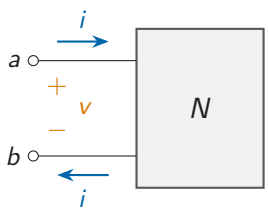
Roadmap

1. **LTI resistive circuits** — what the words mean, and what we gain ✓
2. **A first example** — simplify a circuit step by step ✓
3. **Substitution theorem** — replace a part of a circuit by a source ✓
4. **One-ports and equivalence** — the same i–v characteristic at the port
5. **Series and parallel** — combine resistors and sources
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Ports and one-ports

Port, one-port

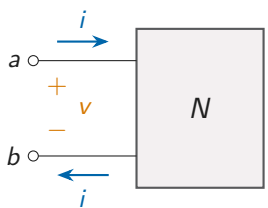
A pair of terminals is a port if the current entering one terminal equals the current leaving the other. A one-port is a circuit, or a circuit element, with one pair of terminals accessible from outside.



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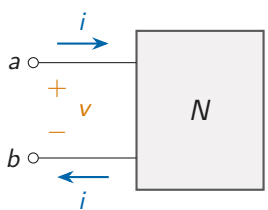


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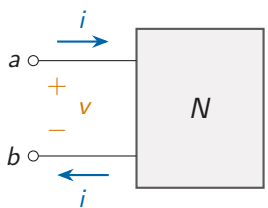


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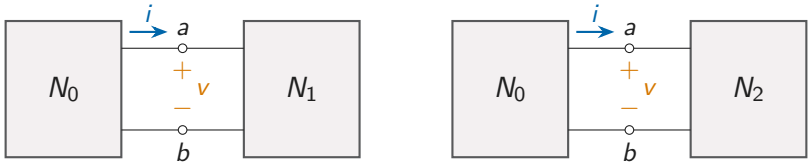


- A one-port is always a port: KCL on a closed surface around N.
- i enters the $+$ terminal of v.
- The relation between v and i is the i–v characteristic, e.g. $v = Ri$ for a resistor.

Equivalent one-ports

Equivalent one-ports

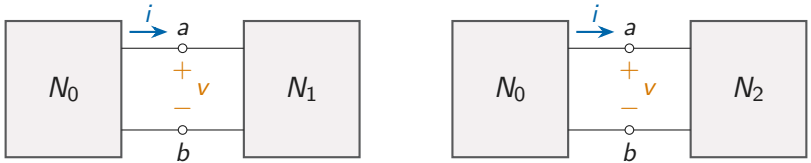
$N_1 \equiv N_2$ if they have the same i–v characteristic at their terminals.



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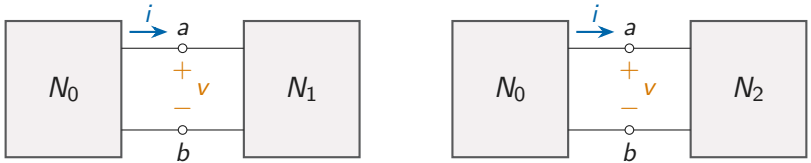


For any N_0: every branch voltage and current inside N_0 is the same in both.

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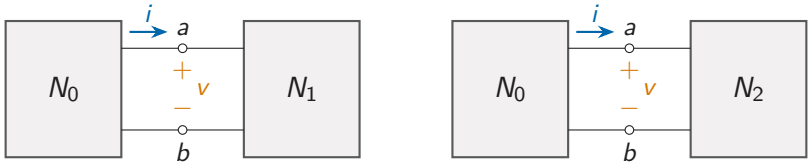
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Why: the equations are those of N_0 plus the i–v characteristic at the port.

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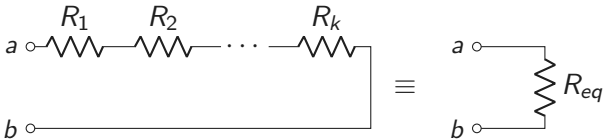
Inside N_1 and N_2 the branch values may differ.

Roadmap

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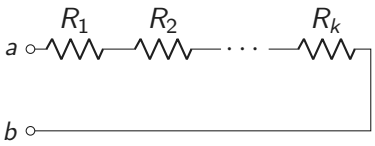
Series connection

Same current i through every element (KCL); the voltages add (KVL).

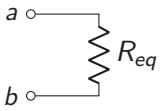


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$\equiv$

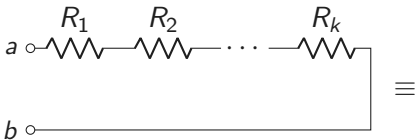


$$R_{eq} = R_1 + R_2 + \cdots + R_k$$

$$v = R_1 i + R_2 i + \cdots + R_k i = R_{eq} i$$

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Same current i through every element (KCL); the voltages add (KVL).

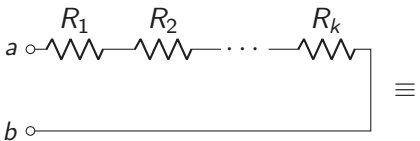


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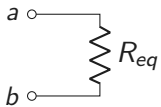
Voltage divider: the share of R_j is $v_j = \dfrac{R_j}{R_{eq}}\, v$

Series connection

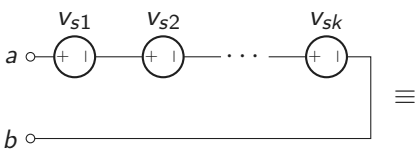
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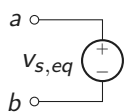
$\equiv$



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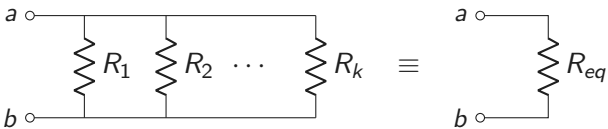


$$v_{s,eq} = v_{s1} + v_{s2} + \cdots + v_{sk}$$

All $+$ signs towards a; a source turned the other way enters with $-$.

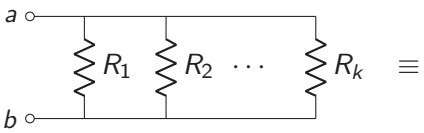
Parallel connection

Same voltage v across every element (KVL); the currents add (KCL).



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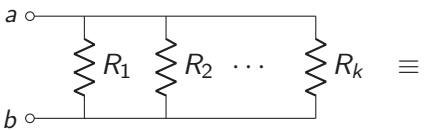


$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_k}$$

With $G_j = 1/R_j$: $\quad i = G_1 v + \cdots + G_k v = G_{eq} v, \quad G_{eq} = G_1 + \cdots + G_k$

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Same voltage v across every element (KVL); the currents add (KCL).

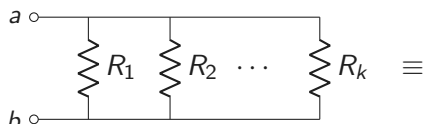


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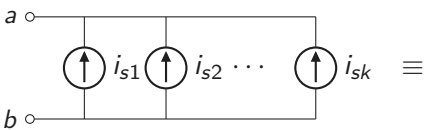
Current divider: $i_j = \dfrac{G_j}{G_{eq}}\, i.$ Two resistors: $R_{eq} = \dfrac{R_1 R_2}{R_1 + R_2}$

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Same voltage v across every element (KVL); the currents add (KCL).



$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_k}$$



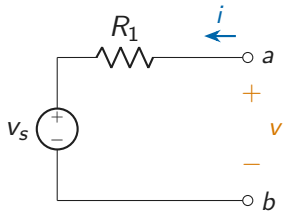
$$i_{s,eq} = i_{s1} + i_{s2} + \cdots + i_{sk}$$

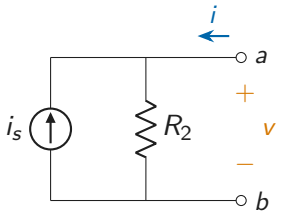
All arrows towards a; a source turned the other way enters with $-$.

Roadmap

1. **LTI resistive circuits** — what the words mean, and what we gain ✓
2. **A first example** — simplify a circuit step by step ✓
3. **Substitution theorem** — replace a part of a circuit by a source ✓
4. **One-ports and equivalence** — the same i–v characteristic at the port ✓
5. **Series and parallel** — combine resistors and sources ✓
6. **One source and one resistor** — exchange a voltage source and a current source

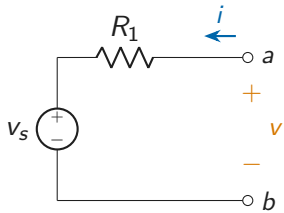
One source and one resistor

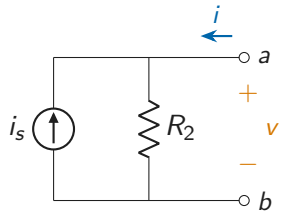




When do these two one-ports have the same i–v characteristic?

One source and one resistor

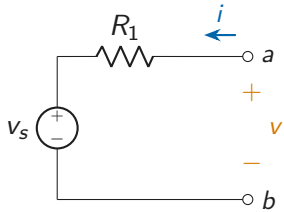




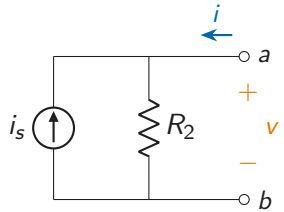
$$v = R_1 i + v_s$$

KVL around the loop: R_1 carries i.

One source and one resistor



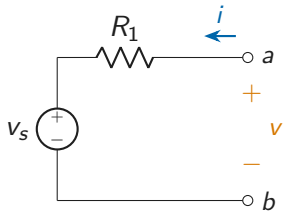
$$v = R_1 i + v_s$$



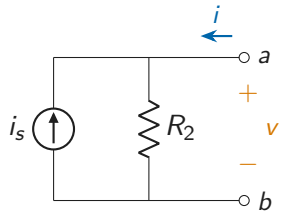
$$v = R_2(i + i_s) = R_2 i + R_2 i_s$$

i and i_s both flow into the top node: R_2 carries $i + i_s$.

One source and one resistor



$$v = R_1 i + v_s$$



$$v = R_2(i + i_s) = R_2 i + R_2 i_s$$

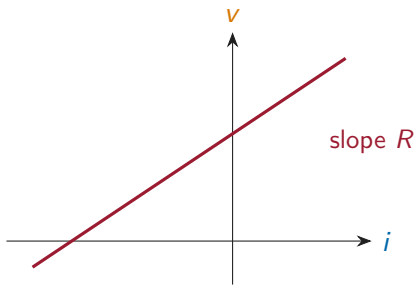
Both are straight lines in the i–v plane: equal slopes and equal intercepts.

Source transformation

Source transformation

The two one-ports are equivalent if

$$R_1 = R_2 = R \qquad \text{and} \qquad v_s = R\, i_s.$$



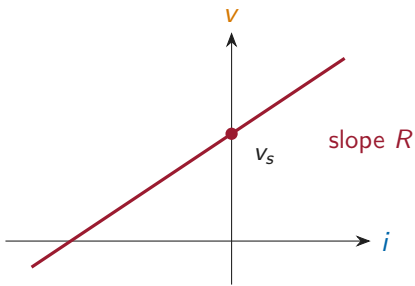
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Open circuit ($i = 0$): $v = v_s$.



Source transformation

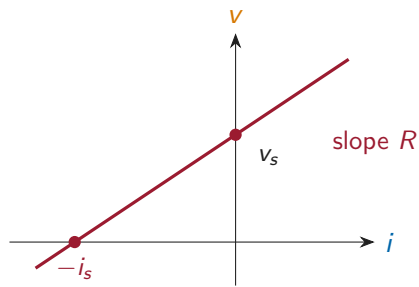
Source transformation

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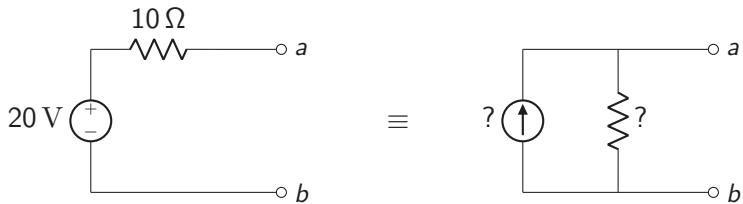
$$R_1 = R_2 = R \qquad \text{and} \qquad v_s = R\, i_s.$$

Open circuit ($i = 0$): $v = v_s$.

Short circuit ($v = 0$): $i = -i_s$.

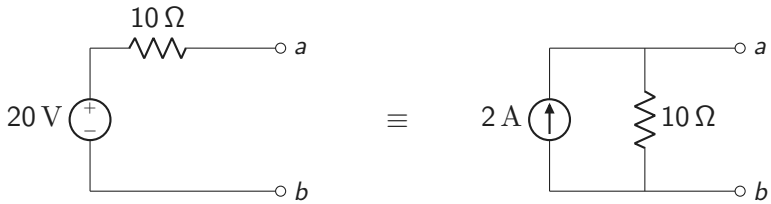


Example: a source transformation



Find the current source and the resistor.

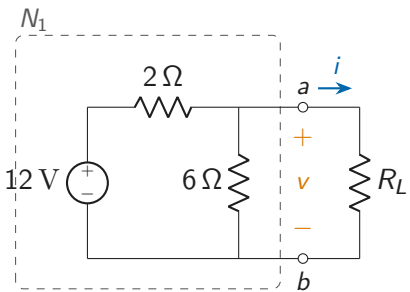
Example: a source transformation



$$R = 10\,\Omega, \qquad i_s = v_s/R = 20/10 = 2\,\mathrm{A}$$

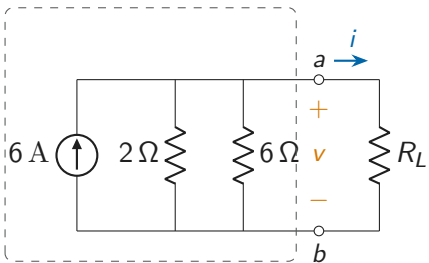
Both: open-circuit voltage $20\,\mathrm{V}$, short-circuit current $2\,\mathrm{A}$ (out of a).

The first example, again



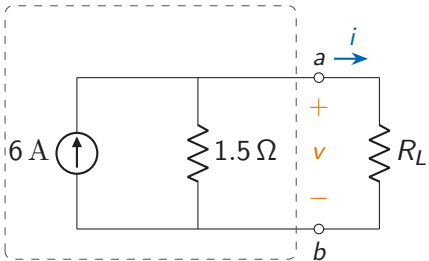
Now simplify only the part left of a–b; R_L stays.

The first example, again



Step 1. Source transformation:
$12\,\mathrm{V}$ with $2\,\Omega \rightarrow 6\,\mathrm{A}$ with $2\,\Omega$ $(12 = 2 \times 6)$

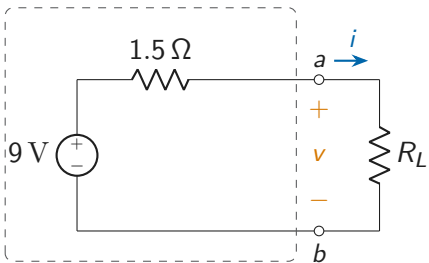
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Step 1. Source transformation:
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The first example, again

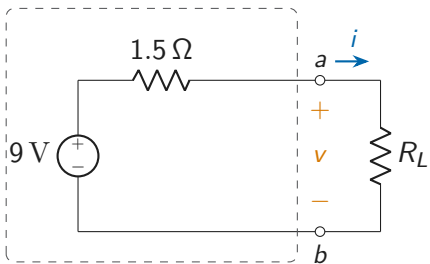


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$6\,\mathrm{A}$ with $1.5\,\Omega \rightarrow 9\,\mathrm{V}$ with $1.5\,\Omega$
$(9 = 1.5 \times 6)$

The first example, again



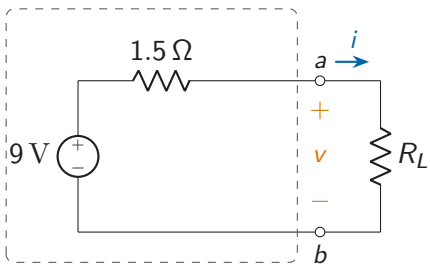
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Same $9\,\mathrm{V}$ and $1.5\,\Omega$, now by transforming the circuit.

The first example, again



Step 1. Source transformation:
$12\,\mathrm{V}$ with $2\,\Omega \to 6\,\mathrm{A}$ with $2\,\Omega$ $(12 = 2 \times 6)$

Step 2. Parallel: $2\,\Omega \parallel 6\,\Omega = 1.5\,\Omega$

Step 3. Source transformation:
$6\,\mathrm{A}$ with $1.5\,\Omega \to 9\,\mathrm{V}$ with $1.5\,\Omega$
$(9 = 1.5 \times 6)$

Same $9\,\mathrm{V}$ and $1.5\,\Omega$, now by transforming the circuit.

Careful: inside, the values differ; for $R_L = 3\,\Omega$ the $12\,\mathrm{V}$ source carries $3\,\mathrm{A}$, the $9\,\mathrm{V}$ source $2\,\mathrm{A}$.

Solving LTI resistive circuits

- Series and parallel connections and source transformations simplify many circuits, but not all.

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  1. node voltage analysis: the $n-1$ node voltages are the unknowns;
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  1. node voltage analysis: the $n-1$ node voltages are the unknowns;
  2. mesh current analysis: the $b-n+1$ mesh currents are the unknowns.
- Both start from the matrices met earlier (A, M), now with linear element equations.

Summary

- LTI resistive: apart from the independent sources (external inputs), every element is linear, time-invariant and memoryless, so all equations are linear and algebraic.
- Substitution: replace N_2 by a voltage source $v_s = v$ or a current source $i_s = i$; if each circuit has a unique solution, nothing changes inside N_1.
- Equivalent one-ports: the same i–v characteristic; the rest cannot tell them apart.
- Series: $R_{eq} = \sum R_j$, $v_{s,eq} = \sum v_{sj}$. Parallel: $G_{eq} = \sum G_j$, $i_{s,eq} = \sum i_{sj}$.
- Source transformation: v_s with series $R \equiv i_s = v_s/R$ with parallel R.